

4.1

Chceme najít koeficienty a, b, c roviny z , aby byly čtverce co nejmenší.

$$(x_i, y_i, z_i): (0, 0, 1), (1, 0, 1), (0, 1, 1), (1, 1, 2), (1, -1, 0), (-1, 1, 1)$$

Pro každý bod by mělo platit $ax_i + by_i + cz_i = z_i \Rightarrow$

$$Ax = b \text{ kde } x = (a, b, c)^T$$

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \text{neboli cílové hodnoty}$$

$$A^T A x = A^T b$$

$$A^T A = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 2 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 2 \\ 0 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

$$A^T b = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 2 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

Řešení soustavy:

$$4a + 2c = 2 \quad \dots \quad 2a + c = 1 \quad \dots \quad c = 1 - 2a$$

$$4b + 2c = 4 \quad \dots \quad 2b + c = 2 \quad \dots \quad 2b + 1 - 2a = 2 \quad \dots \quad b - a = \frac{1}{2}$$

$$2a + 2b + 6c = 6 \quad \dots \quad a + b + 3c = 3 \quad \dots \quad a + b + 3 - 6a = 3 \quad \dots \quad b = 5a$$

$$5a - a = \frac{1}{2} \quad a = \frac{1}{8} \quad b = \frac{5}{8} \quad c = 1 - \frac{2}{8} = \frac{3}{4}$$

$$\underline{\underline{z = \frac{1}{8}x + \frac{5}{8}y + \frac{3}{4}}}$$